RUTURAJ DIXIT

Data Scientist · AI-Native Operator · B2B Product Analytics & Experimentation

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SUMMARY

Data scientist who has shipped AI for global B2B enterprises (Tokyo Electron, Lamb Weston, Stellantis) and founded an AI-native product (PrimeMarket AI). Owns the insight → hypothesis → experiment → decision loop: KPI frameworks, funnels, A/B tests, and roadmap-shaping insight from messy signals. AI-native daily operator (LLMs, agents, RAG). M.S. Data Science @ Pace; AWS-certified ML Engineer; vision-language model researcher at the Pace AI Lab.

CORE COMPETENCIES

Product Analytics: KPI frameworks · onboarding/activation/retention funnels · time-to-value · north-star metrics · cohort analysis
Experimentation: A/B testing · power analysis · CUPED · diff-in-diff · pre/post & switchback designs · guardrail metrics
AI/ML & HCI: LLMs, RAG (LangChain) · agentic workflows · model eval (AUC, F1, calibration) · trust & quality metrics
Stack: Expert SQL · Python (Pandas, NumPy, scikit-learn, LightGBM) · PySpark · Airflow · Snowflake · Power BI

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant & AI Lab Researcher | Pace University · New York, NY *Oct 2025 – Present*

- Selected by faculty (Profs. Sara Falcone, Paul Benjamin) to instruct/grade Python, ML, and AI Systems — coaching grads on RAG pipelines, NLP, ML evaluation, and data mining.
- Pace AI Lab researcher on the ADAPT architecture, fine-tuning frontier vision-language and depth-foundation models (Meta SAM 3D, Depth Anything v3, VGGT) — directly relevant to measuring human-AI interaction quality and trust.
- AI-native daily: ChatGPT, Claude, Cursor, agentic tooling as throughput multipliers across research, code, and writing.

Founder & Data Scientist | PrimeMarket AI (primemarketai.com) · Independent *2024 – Present*

- Defined product north-star (forecast directional accuracy on held-out intraday windows) and a 'time-to-first-signal' activation metric — same time-to-value framing OpenAI Work uses.
- Built experimentation harness from scratch: rolling-window backtests, walk-forward validation, champion/challenger across XGBoost, LSTM/RNN, and transformer forecasters; promoted only models passing Sharpe/drawdown guardrails.
- Instrumented full funnel (landing → signup → first forecast → repeat session); used cohort retention curves to prioritize highest-leverage UX fixes; doubled D7 repeat-session rate after one onboarding change.
- Stack: Python, FastAPI, PyTorch, automated feature engineering, Docker, AWS — deployed and monitored as a live service.

AI / ML Engineer — Data Science & Analytics | Tata Consultancy Services · Pune, India *Oct 2022 – Aug 2024*

Embedded data scientist for global B2B clients (Tokyo Electron — semiconductor; Lamb Weston — CPG; Stellantis — automotive).

- Defined KPI frameworks and end-to-end funnels for client AI rollouts — adoption, accuracy SLAs, throughput, and quality guardrails — so leadership had a single trusted dashboard.
- Ran controlled A/B comparisons across 6+ model variants (YOLO, Detectron2, ResNet, X3D, ANN) using AUC, F1, calibration — shipped versions improved operator workflow efficiency by up to 40%.
- Built SQL-driven analytical pipelines on multi-million-row enterprise data to surface drift, cohort regressions, and edge-case failures — feeding insights back into product iteration.
- Drove instrumentation upgrades with engineering: standardized event schemas across OCR (AWS Textract) and CV pipelines for near-real-time quality, latency, and human-override metrics.
- Delivered production-grade systems achieving 80–90% accuracy across CV/OCR; reduced manual review effort and processing time at every client.

SELECTED PROJECTS & RESEARCH

Pattern Recognition in Financial Charts — Peer-reviewed (ResearchGate)

- Image-segmentation method for detecting Fair Value Gaps, trendlines, and harmonic patterns in market visuals.

Antler Hackathon — ChatGPT API integration with Pincaster

- Shipped a working LLM-powered conversational layer over a real product surface in under 72 hours.

EDUCATION

M.S. in Data Science | Pace University · New York, NY *Expected May 2026*

Coursework: Statistical Inference, Machine Learning, Big Data, NLP, Data Mining, Experimentation.

B.Tech in Electronics Engineering | Shivaji University · India *July 2022*

CERTIFICATIONS

AWS Certified Machine Learning Engineer – Associate · Databricks Fundamentals Certified · PCAP: Programming Essentials in Python